



Nirbhay Singh

Machine Learning | GenAI | LLMOps
MLOps | Vertex AI | SageMaker

CONTACT

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CORE SKILLS

AI / ML / GENAI

Vertex AI, SageMaker, LLMs (GPT-4, Claude, Gemini), RAG, Prompt Engineering, Agentic AI, LangChain, Vector DBs

MLOps & LLMOPS

Model Registry, Feature Store, ML Pipelines, A/B Testing, Model Monitoring, Data Drift Detection

DATA & ANALYTICS

BigQuery, Redshift, EMR, Spark, Enterprise Search, Data Pipelines, ETL

CLOUD

GCP, AWS, Azure, Multi-Cloud Architecture

INFRASTRUCTURE

Terraform, Kubernetes, Docker, CI/CD, GitOps

LANGUAGES

Python, Bash, SQL, YAML, HCL

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

Nirbhay Singh

Cloud & AI Architect | ML/AI & Generative AI Specialist

11+

Years Experience

14+

Certifications

EUR 4M+

Cloud Savings

1,000+

Containers Orchestrated

60%

AI Cost Reduction

PROFESSIONAL SUMMARY

Cloud & AI Architect who has spent 11+ years turning ML research into production systems on AWS, GCP, and Azure. At Bosch Europe, leads LLM and Agentic AI integration into enterprise search and document intelligence, running on Vertex AI, BigQuery, and Cloud Run and adopted across multiple business units. Builds the MLOps scaffolding underneath that work -- training pipelines, evaluation, model registries -- and open-sources the cost-control layer: TokenMeter, which cut LLM spend 60% through provider-aware routing. Google Cloud Certified Professional Machine Learning Engineer and Generative AI Leader.

KEY ACHIEVEMENTS

- Architected a RAG-based enterprise search platform on Vertex AI, grounding retrieval in Bosch's internal document corpus and rolling it out to 15+ business units
- Built end-to-end MLOps pipelines -- training, evaluation, registry, deployment, drift monitoring -- with retraining triggered automatically on data-drift thresholds
- Open-sourced TokenMeter, a cost-intelligence layer that meters tokens across OpenAI, Anthropic, and Google and routes by task complexity, cutting LLM spend 60%
- Ran production ML inference on GKE, EKS, and AKS with canary rollouts and model versioning, keeping every deployment reversible at scale
- Built ShieldIaC, an IaC scanner pairing 100+ static rules with GPT-4.1-generated fix suggestions across 9 compliance frameworks

PROFESSIONAL EXPERIENCE

- Cloud & AI Architect** Dec 2022 - Present
 - Robert Bosch (Bosch Polska)** Warsaw, Poland
 - Led GenAI adoption across Bosch Europe, integrating LLMs and Agentic AI workflows into enterprise search on Vertex AI, BigQuery, and Cloud Run
 - Built a RAG-based document intelligence platform indexing 500K+ documents with semantic search, deployed across 15+ business units
 - Established the LLMOps framework Bosch now runs on: prompt versioning, evaluation pipelines, cost tracking, and production guardrails
 - Set AI/ML architecture standards for Bosch Poland's joint initiative with Google Cloud and AWS
 - Drove GPU scheduling and preemptible-instance strategy for AI/ML workloads, part of a EUR 4M+ FinOps programme
- Senior DevOps Consultant (ML Platform)** Jun 2021 - Nov 2022
 - McAfee**
 - Ran ML inference on GKE and EKS with autoscaling and canary rollouts for real-time serving
 - Built drift detection and alerting on Prometheus and Grafana to catch model degradation before it hit production
 - Automated ML infrastructure with Terraform, cutting deployment time from days to hours
- Senior DevOps Engineer** Dec 2018 - Jun 2021
 - Deloitte**
 - Built data and AI platforms on SageMaker, EMR, and Redshift for model training and Fortune 500-scale batch inference
 - Orchestrated 1,000+ containers on AWS Batch and EKS with Spot Instances for distributed ML training
 - Cut ML infrastructure cost \$45,000 through Spot Instance optimisation and GPU right-sizing

Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

PTC

- Designed fault-tolerant AWS architectures (EC2, RDS, S3, VPC) for data ingestion pipelines and early ML workloads
- Led migration of 11,000+ servers to the cloud, unlocking cloud-native ML and analytics tooling for data teams

System Engineer

Mar 2013 - Jul 2014

IBM

- Handled P1/P2 incident response and root cause analysis for production enterprise systems

OPEN-SOURCE PROJECTS

TokenMeter -- Cost Intelligence Layer for LLM Apps

>> github.com/nirbhays/tokenmeter

Cuts AI spend up to 60% with a one-line Python integration that tracks cost, latency, and tokens across OpenAI, Anthropic, and Google, then routes by task complexity. <5ms overhead.

STACK: Python, OpenAI API, Anthropic API, Google AI, Smart Routing, PyPI

ShieldIaC -- AI-Powered IaC Security Scanner

>> github.com/nirbhays/shieldiac

Flags Terraform and CloudFormation misconfigurations against 100+ rules across 9 compliance frameworks (CIS, SOC2, HIPAA, PCI-DSS, NIST), then suggests fixes via GPT-4.1.

STACK: Python, Terraform, CloudFormation, GPT-4.1, GitHub Actions

InfraCents -- Terraform Cost Estimates on Every PR

>> github.com/nirbhays/infracents

Posts real-time AWS+GCP cost estimates directly on pull requests by parsing Terraform diffs against live pricing APIs -- zero configuration.

STACK: Python, Next.js, Terraform, GitHub App, AWS/GCP Pricing APIs

CERTIFICATIONS & CREDENTIALS

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|---|---|
| ■ Google Professional Machine Learning Engineer (Valid Dec 2027) | ■ Google Generative AI Leader (Valid Aug 2028) |
| ■ Google Professional Cloud Architect (Valid Jun 2026) | ■ Google Professional Cloud DevOps Engineer (Valid Dec 2027) |
| ■ Google Associate Cloud Engineer (Aug 2023) | ■ AWS Solutions Architect - Professional (Feb 2020) |
| ■ AWS Solutions Architect - Associate (Jan 2021) | ■ CKAD - Certified Kubernetes Application Developer (Valid Sep 2026) |
| ■ HashiCorp Terraform Associate | ■ MCSA - Microsoft Certified Solutions Associate (Apr 2015) |

TECHNICAL WRITING & THOUGHT LEADERSHIP

- "Our AI Bill Was \$4,800 Last Month -- Nobody Knew Why" | "How I Built an AI-Powered IaC Security Scanner"
Medium deep-dives into building TokenMeter (LLM cost intelligence) and ShieldIaC (AI-powered security scanning).
<https://medium.com/@nirbhaysingh1>